

ET AI HACKATHON 2.0

Vayu Console

From pollution signals to defensible action

Urban Air Quality Decision Intelligence

Technical Document

A detailed account of the system, its evidence,
and the claims it deliberately refuses to make.

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vayu-console-web.vercel.app/console • github.com/Pranavsingh431/vayu_console

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Every quantitative claim in this document is traceable to the repository or to the live API. Where a result has not been measured, it says so. Section 18 is the honest list of what the system cannot do, and it is not short.

1 Executive Summary

Delhi has no shortage of air quality dashboards. Every one of them answers the same question: how bad is the air right now. That is not the question a municipal officer has to answer at 2 a.m. during a severe episode. Their question is what to *do*, and how to defend that decision in a review meeting the following morning.

Vayu Console is a decision support system for that gap. It takes an observed pollution episode at a monitoring station and runs it through four stages: **situation, evidence, decision, explanation**. It evaluates three independent hypotheses for what may be contributing, rates the evidence for each one against observations that are, where possible, collected independently of the monitor being explained, applies deterministic policy rules to that evidence, and exposes the complete audit trace behind every recommendation it produces.

The defining design decision is that the system is allowed to return “insufficient evidence” and recommend nothing at all.

That is not an error path or an unhandled case. It is a first-class output with its own visual treatment and its own copy, because the alternative is worse: a system that always answers will, on the many station-hours where the required observations are missing, produce an answer indistinguishable from one that is grounded. An officer cannot tell the two apart. That is not a gap in a user interface, it is a system that lies at a known rate.

Three things are worth a technical reader’s attention before the detail.

1. **This is source-contribution screening, not source apportionment.** Delhi’s regulatory network measures mass concentration, not chemical composition. Without speciation there is no ground truth for attribution, so any percentage this system emitted would be invented. It emits none. There is no pie chart, no stacked bar, and no $\mu\text{g}/\text{m}^3$ figure attributed to any source anywhere in the product.
2. **A supervised source classifier was built, scored 98.6% accuracy, and was deleted.** Its labels were generated by a rule over its own input features, so it had learned the rule rather than the atmosphere. The instructive part is that every instrument we could have pointed at it for honesty — SHAP, conformal prediction, calibration curves — would have corroborated the fabrication rather than exposed it. Section 10 sets out the measurement and the argument.
3. **One validation has been run to completion.** The COVID-19 lockdown calibrated and accepted the traffic module at a likelihood ratio of 2.11, which is “weak” on the Kass & Raftery scale, and the module is capped in code because of it. The other declared tests are pending, the engine reports them as pending at `/evidence/history`, and the console shows them as pending. Section 11 does not round any of them up.

2 Problem Statement

A municipal air quality officer in Delhi during a severe episode has access to a real-time $\text{PM}_{2.5}$ reading from a network of monitoring stations, and to public AQI dashboards that render that reading with a colour and a category label. What they do not have is any structured answer to the questions that actually determine their next action:

- Why is this happening, and what evidence supports each candidate explanation?
- Which of the available interventions is proportionate to that evidence?
- How confident should anyone be in that judgement?
- If challenged tomorrow, what exactly is the reasoning, and where are its weaknesses?

None of these are measurement problems. The measurement exists. They are reasoning problems, and at present they are solved by an officer manually correlating fragmented signals — a station reading here, a news report of stubble burning there, a sense of the wind — under time pressure, with no record of how the conclusion was reached.

The gap is not between data and dashboards. It is between a reading and a decision an officer can defend.

3 Problem Analysis and Motivation

The obvious response to this gap is to build a model that predicts the pollution source, attach a confidence score, and present the result. That approach fails for a reason specific to the data, and understanding the reason shaped the entire system.

3.1 There is no ground truth for attribution

Identifying which source produced a given mass of $\text{PM}_{2.5}$ is called *source apportionment*. It is a mature field, and it works by chemical speciation: collecting filter samples, measuring the elemental and organic composition, and solving a receptor model such as Chemical Mass Balance or Positive Matrix Factorisation against known source profiles.

Delhi's regulatory network does not do this. It measures mass concentration ($\text{PM}_{2.5}$, PM_{10}) and a handful of gaseous species (NO_2 , SO_2 , CO , O_3). There is no composition data, and therefore no labelled dataset of “this station-hour was caused by vehicles” anywhere in the archive — because producing one would itself require the apportionment we are trying to perform.

This has a hard consequence. A supervised model needs a target. If no target exists, one must be invented, and any invented target is a function of the same features the model will be given. Section 10 shows what happens when you do that anyway.

3.2 What is available instead

The asymmetry that makes the system possible is that *some* hypotheses have evidence that exists outside the measurement being explained, and some do not.

Satellites observe agricultural fires in Punjab and Haryana independently of any Delhi air quality monitor. That observation is exogenous: it cannot be a restatement of the thing it is being used to explain. Traffic has no such signal — there is no satellite watching Delhi's cars — so its only signature lies in the same chemistry we would be reasoning from. Industry has neither an exogenous signal nor a natural experiment that isolates it.

This asymmetry is not a nuisance to be smoothed over. It is reported as a first-class property called *identification*, separately from evidence strength, because it does not vary hour to hour. It is a statement about what can be known at all.

3.3 Why historical events rather than live data

Live data cannot validate a reasoning system. You produce a recommendation and never find out whether it was right, because the counterfactual is unobservable.

Historical episodes in which a known external intervention changed one input while leaving others in place are the only setting where the logic can be tested. The COVID-19 lockdown drove traffic to approximately zero by government order, which makes $P(\text{NO}_2 \mid \text{vehicles} \approx 0)$ directly observable rather than assumed. No researcher could run that experiment deliberately.

The console therefore replays reconstructed historical incidents, and says so on every screen.

4 Proposed Solution: Vayu Console

Vayu Console implements a four-stage pipeline, and its interface is laid out in that order because the layout *is* the workflow.



Situation presents the measured concentration against the Central Pollution Control Board's published $\text{PM}_{2.5}$ breakpoints. Severity is deliberately *not* derived from evidence strength: evidence says which explanation is plausible, it says nothing about how bad the air is. An officer can cite a measured number against a published threshold; they cannot cite a model output.

Evidence evaluates three hypotheses independently and reports, for each, its strength, the quality of the data underneath it, its identification status, the supporting and contradicting observations, the assumptions in play, and its historical validation record.

Decision applies deterministic rules to the evidence report. The rule engine may read nothing else. If a fact is not in the evidence report, no rule can use it, which is what guarantees that every recommendation traces back to an observation.

Explanation is the Challenge Recommendation interaction: one click on any recommendation exposes the full trace from advice back to observation, including the evidence that argues *against* it.

4.1 The architectural invariant

The Decision Engine consumes the Evidence Report and touches no raw data.

This is enforced structurally rather than by convention. The decision layer’s only input type is `EvidenceReport`. It has no database session and no access to station readings, satellite detections, or meteorology. The consequence is that no rule can reach a recommendation that the evidence does not carry, and every recommendation is reconstructible from a serialisable object.

5 System Architecture

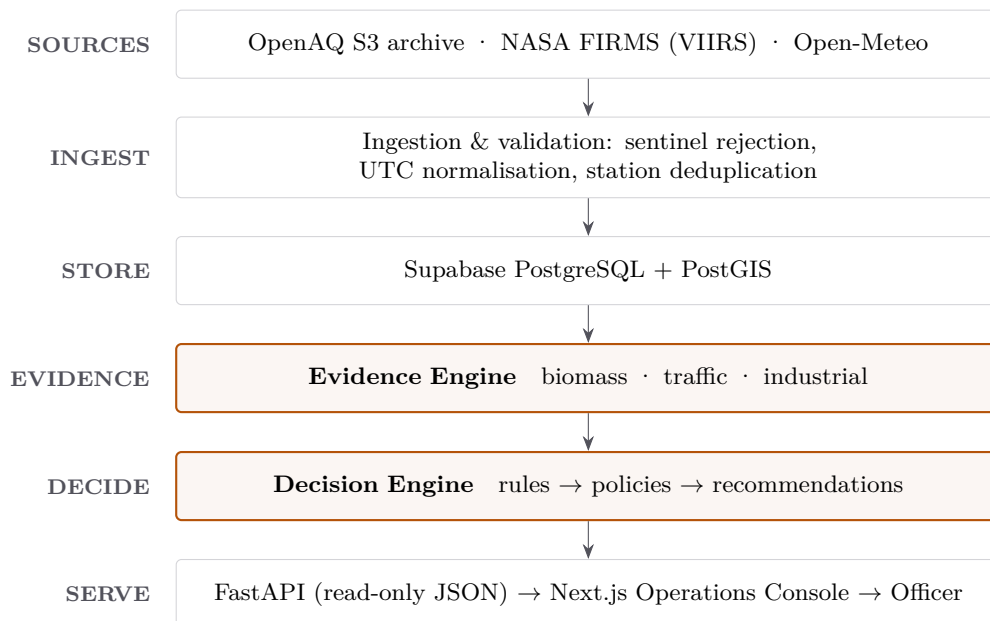


Figure 1. System architecture. The two highlighted layers are where the reasoning happens; everything above them is data handling and everything below is delivery.

The reasoning flow within those two layers, including the branch that produces no recommendation at all:

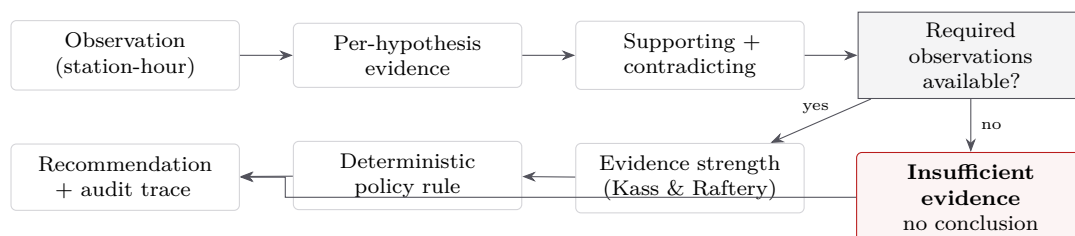


Figure 2. Reasoning flow. The “insufficient evidence” branch terminates in the same audit trace as a recommendation would, so a refusal is as inspectable as an answer.

6 Data Sources and Data Pipeline

Three external sources are ingested. There is no fourth.

Source	Provides	Role in the system
OpenAQ S3 archive	Station readings (CPCB origin)	PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO, O ₃ per station-hour. The situation, and the inputs to the traffic and industrial hypotheses.
NASA FIRMS	VIIRS active fire detections	Latitude, longitude, acquisition time, fire radiative power, confidence. The exogenous signal underneath the biomass hypothesis.
Open-Meteo	Meteorology	Wind speed and direction, boundary layer height, temperature. Turns a fire detection into a transport argument.

A note on provenance, because it is easy to state this wrongly. CPCB is the originating authority for the station readings, but the system does not integrate with `data.gov.in` directly. Readings arrive through the OpenAQ S3 archive, which republishes them. There is no fourth ingestion pipeline, and the architecture diagram does not claim one.

6.1 Ingestion and normalisation

Records pass through a parse stage that performs four jobs before anything reaches the database:

1. **Sentinel rejection.** Discussed in Section 7; the single most consequential validation step.
2. **Timezone normalisation.** The archive publishes IST with an explicit +05:30 offset. Naive timestamps are treated as IST by convention and everything is stored as UTC. Mixing the two silently is the kind of error that shifts an entire diurnal analysis by five and a half hours.
3. **Station deduplication.** One physical station appears under several OpenAQ location identifiers.
4. **Type coercion and range checks.** A concentration that cannot be parsed as a number is dropped with a debug log rather than coerced.

6.2 Coverage

Ninety-six monitoring locations lie within 25 km of Delhi's centre and have data in the archive. Forty-eight of them have a gap somewhere in their year range. Station counts by year are heavily uneven, and one span in particular cannot be reasoned across at all — see Section 7.

7 Data Quality and Engineering Challenges

This section covers the findings that shaped the system more than any modelling decision did. Each was discovered by working with the data rather than by reading its documentation.

7.1 The live API and the archive disagree about coverage

OpenAQ exposes both a live API and an S3 archive. The live API advertises Delhi monitoring locations for which it will not, in fact, return historical data. Building coverage assumptions on the API would have produced a usable dataset far smaller than expected, and the discrepancy would have surfaced late, after the evidence modules were already calibrated against it.

Coverage was therefore probed against S3 directly, which is the only source whose claimed coverage matches what it returns. This is why the ingestion path targets the archive rather than the more convenient API.

7.2 Missing-data sentinels are published as if they were readings

The CPCB feed emits sentinel values in the same field as real measurements. -999 is the classic one. Values near -476300 were observed in CO.

A concentration cannot be negative. Any negative value is therefore either a sentinel or an instrument fault, and neither belongs in a mean. The filter is:

```
if numeric < 0 or numeric in {-999.0, -9999.0}: drop
```

Sentinels are dropped at parse time, not filtered in queries.

The reason is operational and was learned the hard way. A sentinel that reaches the database is a sentinel that some future query will forget to exclude. An earlier phase of this project stored five of them in PM_{2.5} and they had to be excluded by hand at every call site afterwards. Filtering at the boundary means the invariant holds everywhere downstream without anyone having to remember it.

Zero is explicitly *not* treated as a sentinel. It is a real reading, and there is a test asserting so, because the obvious over-eager version of this filter silently deletes valid data.

7.3 One physical station, several location identifiers

The archive assigns multiple OpenAQ location IDs to what is physically a single monitoring station. Left unresolved this inflates apparent network size and double-counts the same instrument's readings into any city-wide aggregate. Deduplication happens at ingest, against a station inventory built from the probe described above.

7.4 The 2023–2024 archive gap is real

The archive contains one Delhi station for 2023 and two for 2024, against fifty or more in adjacent years.

This is not a loading bug. The archive genuinely lacks the data. The system's response is to surface the gap as a card in the historical validation timeline rather than interpolate across it. No trend may be drawn through that span, and a smooth line through it would be a fabrication that looks like an analysis.

7.5 Satellite observation gaps are not non-detections

VIIRS passes over Delhi roughly twice per day, near 12:00–14:00 and 01:00–03:00 IST. Consequently, “no fire detections in the last three hours” is ambiguous between two states that must never be collapsed:

- an **observation gap** — no overpass occurred, so nothing could have been seen; and
- a **true non-detection** — an overpass occurred and found nothing.

The system distinguishes them. An observation gap is reported as missing evidence, never as evidence of absence, and the distinction is surfaced to the officer as an explicit assumption in the Challenge modal. Cloud cover compounds the problem: during heavy winter haze the satellite may under-report precisely when the question is most urgent.

This finding turned out to be load-bearing in an unexpected way, which Section 11 returns to.

8 The Evidence Engine

Three modules implement a common interface. Each receives an `EvidenceContext` — a station-hour's pollutants, meteorology and fire observations — and returns an `EvidenceResult` carrying strength, data quality, identification, supporting and contradicting observations, assumptions, limitations, references and historical validation records.

Modules run independently. One module raising an exception is caught, logged, and converted into an insufficient-evidence result for that hypothesis only, so a defect in one module never costs the officer the evidence the others did produce.

Evidence strengths do not sum to 1, and the interface says so.

The hypotheses are not mutually exclusive. On a Diwali night during stubble season, fire and traffic evidence are both legitimately present. Any visual implying shares — a pie chart, a stacked bar, a normalised score — would reintroduce the apportionment the system exists to refuse. The evidence panel carries the sentence “These do not sum to 100% and are not source shares” permanently, and the end-to-end test suite asserts its presence in the rendered DOM.

8.1 Strength scale

Where a module can compute a likelihood ratio, it is mapped onto Kass & Raftery bands rather than an invented scale:

Folded likelihood ratio	Strength
< 3	Very weak
< 10	Weak
< 30	Moderate
< 100	Strong
≥ 100	Very strong

A ratio below 1 favours the hypothesis’s *absence*, so the value is folded via $\max(\text{LR}, 1/\text{LR})$ and the band then describes the strength of evidence in whichever direction it points. Direction is communicated separately; a strength label alone never implies the hypothesis is true. The interface states “not a probability” wherever a ratio is displayed.

8.2 Fire / biomass module

Identification: strong. FIRMS observes fires from orbit, independently of the air quality monitor being explained. This is the only hypothesis with a genuinely exogenous signal.

The module scores each detection’s influence on the station:

$$\text{influence} = \text{FRP} \times \delta_{\text{dist}} \times \alpha_{\text{wind}} \times \rho_{\text{recency}} \times \mu_{\text{mixing}}$$

with each term a stated modelling choice rather than a fitted parameter:

- **FRP** — fire radiative power, the closest available proxy for emission. Missing FRP falls back to 1.0 rather than 0: a detection with no FRP is still a fire, and zeroing it would silently discard it.
- $\delta_{\text{dist}} = 0.5^{d/150}$ — exponential decay halving every 150 km, chosen to reflect the Punjab-to-Delhi transport distance. Detections beyond 400 km score zero.
- $\alpha_{\text{wind}} = \cos \theta$ for $|\theta| < 90^\circ$, else zero, where θ is the offset from the upwind bearing. A fire downwind contributes nothing; smoke does not travel against the wind. If wind is unknown the term is set to 0.25 rather than assumed aligned, because assuming would manufacture evidence out of ignorance.
- $\rho_{\text{recency}} = 1 - (\text{age}/24)$ over a 24-hour lookback, a deliberately crude stand-in for a transport time the system does not model.
- $\mu_{\text{mixing}} = \min(2, 500/\text{BLH})$ when the boundary layer is below 500 m, else 1. A low boundary layer concentrates whatever arrives into less air.

Total influence across detections maps to a likelihood ratio through a stepped table (influence $\geq 1 \rightarrow 1$, $\geq 10 \rightarrow 3$, $\geq 50 \rightarrow 8$, $\geq 200 \rightarrow 20$, $\geq 1000 \rightarrow 60$).

The kernel is a modelling choice, not a calibrated dispersion model, and the Challenge modal says so on every fire-driven recommendation.

Its parameters were chosen from physical reasoning about transport distance, not fitted to the outcome being explained — fitting them would make the kernel circular with the thing it is supposed to explain.

8.3 Traffic module

Identification: weak. There is no satellite watching Delhi’s cars. The only signature available lies in the same chemistry we would reason from, which is exactly the circularity Section 10 describes.

What rescues the module from being a bare heuristic is the COVID-19 lockdown, which made $P(\text{NO}_2 \mid \text{vehicles} \approx 0)$ observable. The module reasons over the NO_2/SO_2 ratio, on the argument that NO_2 responds strongly to distributed combustion while SO_2 tracks point sources such as coal power. It also considers whether the hour falls in one of Delhi’s two commute peaks (07:00–11:00 and 17:00–21:00 IST).

NO_2 is load-bearing: without it the module returns insufficient evidence rather than reasoning from what remains.

The module is capped at MODERATE strength in code, because the measured likelihood ratio is 2.11.

The cap is the measurement, enforced. No combination of observations can push this module to report strong evidence, because the calibration does not license it.

8.4 Industrial module

Identification: very weak. No natural experiment isolates industry. The COVID lockdown halted traffic, construction and industry simultaneously, so it cannot separate them.

The module therefore carries **no likelihood ratio at all** and reports its absence explicitly rather than borrowing one from a neighbouring hypothesis. It uses SO₂ as a marker for point-source combustion, supported indirectly by the observation that SO₂ held roughly flat (−3.7%) through the lockdown while NO₂ halved (−54.4%), consistent with essential power generation continuing while distributed sources stopped.

SO₂ is measured at only 36 of the 96 stations. At the remaining 60 the module is structurally blind and says so. Absence of an SO₂ sensor is never reported as absence of industrial influence.

9 Scientific Methodology and Positioning

9.1 Screening versus apportionment

	Source apportionment	Source contribution screening (this system)
Output	“Biomass contributed 62 µg/m ³ ”	“Evidence for the biomass hypothesis is moderate”
Method	Receptor modelling (CMB, PMF) over speciated filter samples	Hypothesis-wise evidence from exogenous observations
Requires	Chemical speciation, tracer species, source profiles	Routine monitoring, satellite, meteorology
Ground truth	Measured composition	None available
Validated against	Known source profiles	Natural experiments
Failure mode	Wrong numbers	Wrong ranking, or an honest refusal

The instrumentation gap between these two columns cannot be closed by a model. No amount of statistical machinery converts PM_{2.5} mass into the speciation it was never measured with. The choice was between fabricating percentages and changing the question, and the question was changed.

9.2 What the system may and may not say

Permitted: “Evidence for the biomass hypothesis is moderate.” “Observed wind geometry is consistent with transport from upwind fire activity.” “Required observations are unavailable; no conclusion is drawn.” “This recommendation follows from rule FIRE_002 under policy PUBLIC_HEALTH.”

Prohibited, and absent: any µg/m³ or percentage attributed to a source; any statement that fires *caused* an episode; any probability that a source is responsible; any forecast concentration; any estimate of an intervention’s effect.

These negatives are enforced by test rather than by discipline. The end-to-end suite asserts against the rendered DOM that the words “probability” and “confidence score” never appear, that “do not sum to 100%” always does, and that neither “current situation” nor “current PM_{2.5}” appears anywhere — the last of these guarding against historical reconstructions being presented as live readings.

10 Why the 98.6% Classifier Was Rejected

The project specification called for a supervised classifier predicting pollution source. This section documents why that artefact was built, measured, and then deleted, because the negative result is more informative than the model would have been.

10.1 The circularity

No source labels exist for Delhi. The only labels constructible from available data are rules over the system’s own features — for example, *high NO₂, low SO₂, morning peak, low fire influence* → *Vehicle*.

That is a rule $y = g(x)$. Training $f(x) \approx y = g(x)$ learns g , not the atmosphere.

10.2 The measurement

This was tested rather than assumed. Labels were generated from an NO₂/SO₂ ratio and hour-of-day rule over 13,685 real Delhi station-hours, and a random forest was trained on the rule’s own inputs:

Label distribution (invented by the rule)		
Vehicle	6,396	(46.7%)
Industrial	2,539	(18.6%)
Biomass	4,750	(34.7%)
5-fold cross-validation		
Accuracy	0.9859 ± 0.0035	
Macro F1	0.9851 ± 0.0038	

98.6% accuracy that measures nothing but a random forest’s ability to re-learn an if-statement.

10.3 Why this is dangerous rather than merely useless

The danger is not that the approach fails. It is that it *succeeds*, spectacularly, and in doing so converts every instrument intended for honesty into camouflage:

Instrument	What it reports	What it actually means
Accuracy / F1 / confusion	≈ 0.99, clean diagonal	Rule-reproduction fidelity
SHAP	“NO ₂ dominates the Vehicle class”	NO ₂ is in the rule
Conformal prediction	90% coverage, tight sets	Calibrated coverage <i>of the rule</i>
Reliability diagram	On the diagonal	The rule is self-consistent

Conformal prediction guarantees coverage of y . If y is fiction, it guarantees coverage of fiction, rigorously.

Statistical machinery cannot manufacture the ground truth it requires as input. Applying it here would make the fabrication harder to detect, not easier.

A second, independent objection: the natural experiments that could anchor real labels number roughly five or six. No three-class classifier is estimable at $n \approx 6$.

10.4 What replaced it

Identification status, reported per hypothesis and separately from evidence strength, plus likelihood ratios measured against natural experiments where such an experiment exists and an explicit refusal to report one where it does not. The industrial module carrying no likelihood ratio at all is the clearest expression of this design.

11 Historical Validation

Each module declares, in advance, the real intervention that would reject it. A module that fails its test is removed from the product, not reinterpreted.

One test has been run to completion. The engine reports the others as pending, and the console displays them as pending.

11.1 COVID-19 lockdown, March–April 2020 — ACCEPTED

Traffic went to approximately zero by government order. Measured over Delhi, 1–23 March 2020 against 25 March – 30 April 2020, across 47 stations and 901,160 rows, sentinels excluded:

	Pre-lockdown	Lockdown	Change
NO ₂ (µg/m ³)	32.7	14.9	−54.4%
SO ₂ (µg/m ³)	11.6	11.2	−3.7%
NO ₂ /SO ₂ ratio	2.82	1.34	

NO₂ halved while SO₂ held flat. Power generation remained essential and kept running, which makes this a *differential* intervention rather than a general shutdown, and it is the differential the hypothesis required.

The resulting likelihood ratio is $2.82/1.34 = 2.11$ — “weak” on the Kass & Raftery scale. Three things about that number matter:

1. **It is measured, not chosen.** It exists in code as a constant derived from the ingest, and the traffic module’s MODERATE cap follows from it.
2. **It is an upper bound.** The lockdown also halted construction and industry, so 2.11 over-credits traffic.
3. **The test could have failed.** Had NO₂ not fallen relative to SO₂, the traffic module would have been deleted. A preliminary three-station run gave -51.5% / -6.6% and LR 1.93; the finding replicated at fifteen times the sample.

11.2 Diwali 2019, discriminant validity — PENDING

The question is whether the fire module can absorb firework emissions and misreport them as biomass transport. 1,604 VIIRS detections were present that day, so the confound is real.

There is a strong structural argument that it cannot: VIIRS overpasses Delhi at 12:00–14:00 and 01:00–03:00 IST, entirely outside the 20:00–00:00 firework window, and the module only counts detections that are upwind at 150–250 km. Fireworks in Delhi are neither upwind nor resolvable at that scale.

That is an argument from the satellite’s orbit and resolution. It is not a result, and the test that would confirm it has not been run.

This distinction is worth dwelling on, because an earlier revision of the interface displayed this experiment as VERIFIED while the engine reported it as pending and the human-review panel on the same screen listed it among the pending ones. The console contradicted itself. It was corrected to pending in all four places.

11.3 Odd-Even II, April 2016 — DATA INGESTED, VALIDATION PENDING

The only unconfounded vehicle-restriction window in the record: no stubble burning, no winter inversion. Weak treatment across 11 stations. The data is ingested and the scenario is playable in the console; the analysis has not been run to a conclusion, and it is labelled accordingly rather than presented as a pass.

11.4 Other declared tests

Stubble season versus non-season (biomass) and a dedicated industrial intervention are both declared with failure conditions and both pending. The industrial module’s COVID record is marked *uncertain* rather than accepted, because the lockdown cannot isolate the hypothesis.

12 The Decision Engine

The Decision Engine consumes an Evidence Report and produces a Decision Report. It is deterministic: no model, no language model, no randomness. The same evidence yields the same recommendation every time.

12.1 Layering

Rules map evidence to a *policy*; a policy maps to *recommendations*. Keeping the layers apart means a rule change does not rewrite the advice, and new advice does not rewrite the rules. Five policy families exist: EMERGENCY_RESPONSE, PUBLIC_HEALTH, TRAFFIC_MANAGEMENT, MONITORING, NO_ACTION.

12.2 Two invariants

1. **A rule fires on the presence of evidence, never on its absence.** “Biomass evidence is weak” must never trigger a traffic recommendation, because weak biomass evidence may simply mean the satellite did not pass over. Reasoning from absence would have an officer act on a cloudy sky.

2. **A rule never picks between competing hypotheses.** If two are strong, both sets of recommendations are emitted and the conflict is reported. Choosing would be apportionment by the back door.

12.3 The rule set

Four rules are implemented, each carrying a `defensible_because` field explaining why an officer could defend the resulting action, and a `does_not_imply` field stating what the rule explicitly declines to conclude.

Rule	Condition	Policy
FIRE_001	Biomass strong, traffic at most weak	EMERGENCY_RESPONSE
FIRE_002	Biomass moderate (and not strong)	PUBLIC_HEALTH
TRAFFIC_001	Traffic moderate <i>and</i> hour in a commute peak	TRAFFIC_MANAGEMENT
INDUSTRIAL_001	Industrial at least weak, with SO ₂ actually measured	MONITORING

There is deliberately **no** rule for “traffic strong outside a commute peak”. The traffic module caps at moderate by construction, and elevated NO₂ at 23:00 IST is not a commute profile — the module reports that hour as *contradicting* evidence. A rule recommending rush-hour enforcement at midnight would not be defensible, so it does not exist.

The industrial rules are equally constrained. The only industrial recommendation in the catalogue is REC_INDUSTRIAL_REVIEW, priced INFORMATIONAL, whose action text reads: “*Flag the elevated SO₂ reading for analyst review against known upwind point sources. Do not dispatch inspection on this basis alone.*” Its rationale is that an elevated SO₂ reading is worth a human look but not an enforcement action, because it cannot be attributed to a specific source. A test asserts that no industrial recommendation ever proposes inspection or restriction, so the constraint cannot be relaxed by accident.

12.4 Human review

A decision report is flagged for human review when data quality is poor or unavailable, when multiple hypotheses reach moderate or stronger evidence, when required observations are missing for any hypothesis, when a module’s historical validation is still pending, or when no rule reached an actionable recommendation. Each condition is reported with its reason in the interface, in prose rather than enum identifiers.

13 Challenge Recommendation and Auditability

This is the interaction the product exists for. Any recommendation can be challenged with one click, exposing:

EVIDENCE → RULE → POLICY → RECOMMENDATION

as a step-by-step trace with identifiers, alongside supporting observations, contradicting observations, the confidence note (“how far this can be pushed”), the assumptions that must hold, and the known limitations.

Contradicting evidence is never hidden. An officer needs the counter-argument before someone else in the meeting supplies it.

A worked example, from the Diwali scenario:

Step	Identifier	Detail
Evidence	biomass	moderate, quality high, identification strong
Rule	FIRE_002	Moderate biomass evidence. Defensible because upwind fires are observed but the influence is not decisive; an advisory and closer observation are proportionate, committing the sprinkling fleet is not.
Policy	PUBLIC_HEALTH	
Recommendation	REC_PUBLIC_ADVISORY	Issue a public health advisory for the affected zone

The assumptions listed on that same recommendation include the FRP-as-emissions proxy, the influence kernel being a modelling choice, single-station wind standing in for a 200 km transport path, and satellite overpasses being periodic. The limitations include that the advisory does not reduce emissions and that this is evidence of fire *influence*, not a measure of fire *contribution*.

The point of the interaction is not to convince the officer the system is right. It is to make the reasoning inspectable enough that someone can prove it wrong.

The dialog is keyboard-operable: focus moves into it on open and is restored on close, Escape dismisses it, focus is trapped within it while open, and it scrolls internally so it never exceeds the viewport on a laptop screen.

14 Interface and User Workflow

The console is a single non-scrolling screen. On a control-room display, scrolling the page means losing the number you were looking at.

Reading order, left to right and top to bottom, mirrors the reasoning:

1. **Historical replay banner** — names the mode and the reconstructed incident before anything else on screen.
2. **Situation header** — severity band, observed PM_{2.5}, station, observation time in IST, overall status, and whether human review is required.
3. **Transport geometry** — the station, the wind vector, and upwind fire activity, drawn as SVG.
4. **Evidence** — three hypotheses, expandable to observations and validation records.
5. **Recommended action** — priority, action, rule and policy identifiers, a Challenge affordance per recommendation, human-review reasons, and a reassessment window.
6. **Historical validation** — the natural experiments and their current status.

14.1 Transport geometry, and why it is not a basemap

The visualisation is hand-built SVG: no tile server, no API key, no client-side map library, and nothing that can fail during a demonstration. A street basemap would be decoration. What carries the argument is the geometry — wind arriving from 315°, fires sitting upwind at roughly 200 km.

Fires are drawn as an aggregate in the upwind sector at a representative transport distance, *not* at their true coordinates, because the module reasons over aggregate influence and plotting individual detections would imply a spatial precision the model does not use. The panel footer states this.

14.2 Historical replay labelling

Every scenario is a reconstruction. The console carries a persistent HISTORICAL REPLAY banner naming the incident and its timestamp, uses “Incident situation” rather than “Current situation”, “observed PM_{2.5}” rather than “current PM_{2.5}”, and “analysed” rather than “updated” for the engine run time. The last of these matters: “updated 12:11 pm” beside a 2019 reading reads as a live feed.

14.3 Scenario provenance

Three scenarios are available and one is deliberately absent.

COVID lockdown (15 April 2020) and Odd-Even II (20 April 2016) query real ingested station-hours. Both return **insufficient evidence** for the biomass hypothesis, because fire and weather were never ingested for those windows. That is the system saying it cannot see, and it is the behaviour the project exists to demonstrate.

Diwali 2019 is served from a fixed context held in the API. Its pollutant readings are the recorded Anand Vihar values, but its fire list is a representative set constructed in code rather than a FIRMS query — which is why the station renders as “Anand Vihar (example)”. The engine output is genuine; those particular fire inputs are illustrative. The trade is deliberate: it makes the primary demonstration path survive a cold instance and a fresh clone with nothing ingested.

GRAP is absent because it is a threshold-triggered regulatory regime rather than a dated event, so there is no single station-hour to reconstruct. It is displayed as unavailable with that reason rather than fabricated.

15 Technology Stack

Layer	Choice
Frontend	Next.js 15 (App Router), TypeScript, Tailwind CSS v4, TanStack Query
Visualisation	Hand-built SVG
Backend	Python 3.12, FastAPI, Pydantic v2, SQLAlchemy 2, Alembic
Database	Supabase PostgreSQL with PostGIS
Shared contracts	<code>packages/shared</code> — TypeScript types mirroring the Pydantic schemas
Quality gates	ruff, black, mypy, pytest; ESLint, tsc, Prettier, Playwright
Deployment	Vercel (web), Render, Singapore region (API)

Both applications share types from `packages/shared`, so an API schema change surfaces as a frontend type error rather than a runtime surprise.

There is no language model anywhere in the decision path. This is a design constraint, not an omission: a recommendation an officer may have to defend in a regulatory meeting has to be reproducible, and the Challenge trace is only meaningful if the path is deterministic.

16 Deployment Architecture

The API runs on Render’s Singapore region — closest to Delhi, keeping latency to Supabase and to users sensible — built from a blueprint (`render.yaml`) using the native Python runtime rather than a container, on the basis that it builds faster and is one less moving part. Secrets are declared with `sync: false` so Render prompts for them on first apply and no value is ever stored in the repository.

The web application is deployed on Vercel as a static-prerendered Next.js build.

16.1 A failure mode worth documenting

An empty `CORS_ORIGINS` value disabled the CORS middleware entirely. The symptom was maximally misleading: `/health` continued returning 200 to `curl` while every browser request was blocked and the console rendered blank. Two very different causes — a sleeping service and a healthy service refusing an origin — look identical from the browser.

The service now refuses to boot with a blank `CORS_ORIGINS`, so Render surfaces the misconfiguration at deploy time. Separately, the console’s error state explains *both* plausible causes rather than blaming one, and there is a test asserting that it does.

16.2 Cold starts

The API runs on a free tier that idles. The first request after a period of inactivity can take up to a minute. This is mitigated in the product rather than hidden: the Diwali scenario is served from a fixed context requiring no database, so the primary demonstration path works against a cold instance, and the error state offers an explicit retry.

17 Testing and Reliability

Suite	Scope
pytest	115 tests — evidence modules, aggregator, decision engine, sentinel filtering, config, health
Playwright	34 specs across two viewports (1920×1080, 1440×900)
Static analysis	ruff, black, mypy (strict, 81 source files), ESLint, tsc, Prettier

The end-to-end suite mocks the API at the network boundary by default, so it passes on a laptop with no database and on CI with no secrets, and never wakes a sleeping instance. A suite that requires live infrastructure is a suite nobody runs, and an unrun suite protects nothing. Setting `E2E_LIVE_API=1` runs it against a real backend.

Tests run against a production build rather than the development server. Development overlays and Fast Refresh are not what a judge sees, and a corrupted `.next` development cache once made the console render a permanent skeleton while the API returned 200s.

17.1 Tests that encode product claims

Several assertions exist to protect scientific positioning rather than functionality, which makes them unusually valuable:

- the rendered DOM never contains “probability” or “confidence score”;
- it always contains “do not sum to 100%”;
- it never contains “current situation” or “current PM_{2.5}”;
- the page itself never scrolls at either viewport;
- Escape closes the Challenge dialog and focus moves into it on open;
- an API failure explains both plausible causes rather than one;
- the console always resolves and never renders a permanent skeleton;
- insufficient evidence is never presented as a clean bill of health.

17.2 Reproducibility of submission artefacts

Screenshots in the repository are generated from the running application by a Playwright script rather than captured by hand, so they cannot drift from the build. The presentation deck is generated from a single slide definition that renders the PPTX, the PDF and the markdown content file together, so the three cannot disagree about a figure.

18 Current Limitations

Ranked by how much they should affect an assessment of the work.

18.1 Fundamental

1. **No ground truth for attribution exists.** Nothing here is validated against measured aerosol composition, because none is collected. This is why the system screens rather than apportions, and it is the single most important thing to understand about the project.
2. **Diwali is permanently confounded with stubble burning.** They co-occur every year under the same meteorology. This is structural, not a sampling problem that more data resolves. The system addresses both hypotheses and separates neither.
3. **Traffic is weakly identified and its likelihood ratio is an upper bound.** LR 2.11, capped at moderate in code, and over-crediting traffic because the lockdown also stopped construction and industry.
4. **Industry is very weakly identified and carries no likelihood ratio.** SO₂ reaches only 36 of 96 stations; at the rest the module is blind.

18.2 Methodological

5. **The influence kernel is a modelling choice,** not a calibrated dispersion model, and results are sensitive to it.
6. **FRP is a proxy for emissions.** It measures radiated power, not smoke mass; the relationship is assumed monotonic, not calibrated.
7. **Single-station wind stands in for a 200 km transport path.** In reality wind veers with height and distance.
8. **Severity and evidence are independent by design.** Judges sometimes expect these to be linked. They are not, deliberately.

18.3 Data coverage

9. **The 2023–2024 archive gap.** No trend may be drawn across it.
10. **Sampling is irregular, not hourly.** “Station-hour” is a normalisation, not a native unit of the data.
11. **Live and archived readings are not the same measurement.** The live feed is provisional and later revised.
12. **Cloud cover suppresses fire detection when it matters most,** during heavy winter haze.

18.4 Scope

13. **Delhi only.** The modules are Delhi-calibrated.
14. **Historical replay, not live operation.** Deliberate.
15. **Only one validation completed.** COVID, on the traffic module.
16. **The system does not forecast.** It offers reassessment windows, never predicted concentrations, and cannot estimate the effect of any action it recommends.
17. **Desktop only,** designed for 1440×900 and above. Mobile is not supported and is not claimed to be.

19 Scalability and Future Work

19.1 What ports and what does not

The architecture is city-agnostic: ingestion, validation, the module interface, the Decision Engine and the entire console would run against another city’s data unchanged.

The calibration does not port. Every likelihood ratio is measured against a local natural experiment.

Reusing Delhi’s LR of 2.11 in Mumbai would be precisely the unearned transfer this project refuses elsewhere. Scaling therefore means identifying each city’s natural experiments and re-running the calibration: a data problem, not a configuration change. That is a less impressive claim than one-click multi-city support, and it is the accurate one.

19.2 Not yet built

- **Road-density context** for the traffic hypothesis, giving it a spatial term it currently lacks.
- **Industrial and power-plant proximity**, which would give the industrial module a spatial term and possibly its first real identification.
- **Completing the declared validations** — Diwali discriminant validity, stubble season versus non-season, Odd-Even II.
- **Live ingestion scheduling.** The pipeline supports it; the validation story does not benefit from it, which is why it was not prioritised.
- **Additional cities**, subject to the calibration constraint above.
- **Policy-specific integrations**, such as emitting a GRAP-stage recommendation once a defensible reconstruction of a GRAP episode exists.

19.3 Deliberately excluded

Speciation-based apportionment requires instrumentation Delhi does not have, and no modelling effort substitutes for it. Odd-Even I and III are both confounded — by winter inversion and exemptions, and by peak stubble season respectively — and are listed as pending rather than quietly dropped.

20 Conclusion

Vayu Console addresses a gap that is not about measurement. Delhi has the readings. What it lacks is a reasoning layer between a reading and a decision an officer can defend the next morning.

The system’s central design commitment is negative rather than positive: it will not claim more than the evidence supports. That commitment produced the decisions that most distinguish the work — deleting a 98.6%-accurate classifier because its labels were circular, reporting evidence strengths that deliberately do not sum to one, capping a module in code at the strength its calibration licenses, carrying no likelihood ratio for a hypothesis no experiment isolates, showing an archive gap instead of interpolating across it, distinguishing a satellite observation gap from a non-detection, marking a validation pending when it has not been run, and treating “insufficient evidence” as a designed output rather than a failure.

Most systems are designed to always give an answer. Vayu Console is designed to give an answer only when the evidence can support one, and to show exactly why.

21 Repository and Deployment

Operations Console	https://vayu-console-web.vercel.app/console
API	https://vayu-console-api.onrender.com
health	https://vayu-console-api.onrender.com/health
validation record	https://vayu-console-api.onrender.com/evidence/history
worked example	https://vayu-console-api.onrender.com/evidence/example
OpenAPI docs	https://vayu-console-api.onrender.com/docs
Repository	https://github.com/Pranavsingh431/vayu_console
Release tag	v1.0.0-hackathon

21.1 Where to look in the repository

docs/research/inference.md	The classifier rejection argument in full
docs/research/scientific-limitations.md	Exhaustive limitations
apps/api/app/evidence/	The three modules and the aggregator
apps/api/app/decision/	Rules, policies, recommendation catalogue
submission/judge-faq.md	Answers to the hard questions
submission/final-presentation.pdf	The pitch deck

Team [singhpranav431](#) • ET AI Hackathon 2.0 • Every figure in this document is traceable to the repository or the live API. Claims that could not be verified were excluded rather than softened.